Abstract

This project evaluated single-cell RNA-sequencing data from Drosophila melanogaster brains exposed to cocaine or sucrose across female and male biological groups, following the Project 2 workflow. Eight 10x-style datasets comprising 88,991 cells and 17,481 genes were loaded and merged. Quality-control filtering removed 11 low-feature cells and 4,340 genes detected in fewer than three cells, leaving 88,980 cells and 13,141 genes. After log-normalization, selection of 2,000 highly variable genes, regression of total counts and mitochondrial percentage, PCA, neighborhood construction and Leiden clustering at resolution 0.8, the supplied analysis produced 14 visible clusters rather than the approximately 36 clusters targeted by the guide. Cluster marker analysis generated distinct marker profiles, but canonical neurotransmitter/cell-type markers could not be mapped because the working gene identifiers were Dmel_-prefixed IDs and the notebook did not contain a gene-symbol mapping. Consequently, definitive assignment of clusters to Kenyon cells, glia, dopaminergic neurons, or other neurotransmitter classes was not supported by the supplied analysis. Whole-sex differential expression identified 104 significant genes in males (88 up, 16 down) and 183 in females (162 up, 21 down), using $|\log2FC|>1$ and adjusted $P<0.05$. Thus, the present pipeline does not reproduce the strong male-biased DEG count reported in the reference study. Only 37 significant genes overlapped between sexes. Male-focused pathway plots highlighted neuronal, membrane, metabolic and terminal-bouton-associated Gene Ontology terms, but none of the displayed terms reached an adjusted P-value below 0.05. Overall, the dataset supports substantial cocaine-associated transcriptional remodeling and sex-specificity, but the current pipeline has important technical limitations—especially mitochondrial/ribosomal QC annotation, cluster resolution, gene-symbol mapping, replicate-aware inference, and pathway significance—that must be addressed before making strong cell-type-specific biological claims.

1. Introduction

Cocaine is a potent psychostimulant that alters monoaminergic signaling by inhibiting neurotransmitter reuptake. This report focuses the Drosophila brain as a tractable system for resolving cell-population-specific responses to acute cocaine exposure and emphasizes three analytical questions: construction of a cellular atlas, identification of cell-type-specific cocaine responses, and characterization of sexual dimorphism and pathway remodeling.

The reference study analyzed duplicate female and male samples following sucrose or cocaine-supplemented consumption. Its integrated single-cell analysis yielded 36 stable clusters and reported a substantially larger transcriptional response in males than females, with particularly strong responses in glial and Kenyon-cell populations. The present report evaluates how closely the supplied Scanpy implementation reproduces those biological and computational features.

The project specifically examines sexual dimorphism by comparing female and male flies under cocaine and sucrose exposure. The principal analytical questions are whether a reproducible cellular atlas can be constructed, which cell populations show the strongest cocaine-associated transcriptional changes, whether responses differ between sexes, and which biological pathways are associated with cocaine-responsive genes.

2. Research Questions and Objectives

- Can the supplied Scanpy workflow reproduce a high-resolution Drosophila brain cell atlas approaching the reference 36-cluster solution?

- Which cell populations appear transcriptionally responsive to cocaine, and can canonical neuronal/glial/neurotransmitter markers support annotation?
- Do male and female flies show different magnitudes and identities of cocaine-responsive genes?
- Which biological processes or cellular components are represented among cocaine-associated genes?

3. Materials and Methods

3.1 Dataset and experimental design

The analysis notebook loaded eight datasets: Female_Cocaine_1/2, Female_Sucrose_1/2, Male_Cocaine_1/2, and Male_Sucrose_1/2. Cell counts ranged from 9,072 to 13,193 cells per sample. The merged object contained 88,991 cells × 17,481 genes before QC.

Sample	Condition	Cells
Female_Cocaine_1	Female_Cocaine	13,072
Female_Cocaine_2	Female_Cocaine	9,367
Female_Sucrose_1	Female_Sucrose	9,072
Female_Sucrose_2	Female_Sucrose	11,693
Male_Cocaine_1	Male_Cocaine	10,437
Male_Cocaine_2	Male_Cocaine	11,124
Male_Sucrose_1	Male_Sucrose	13,193
Male_Sucrose_2	Male_Sucrose	11,033

3.2 Quality control

The notebook attempted to annotate mitochondrial genes using the prefix 'mt:' and ribosomal genes using 'RPS/RPL/rps/rpl', then calculated n_genes_by_counts, total_counts, mitochondrial percentage, and ribosomal percentage. Cells with fewer than 200 detected genes and genes detected in fewer than three cells were filtered; cells with mitochondrial percentage ≥10% were also intended to be removed.

A critical QC caveat is that the notebook's QC output reported 0.0% mitochondrial and 0.0% ribosomal counts for all 88,991 cells. This is best interpreted as a feature-identification failure rather than evidence that the fly brain contains no mitochondrial or ribosomal transcripts, because the feature identifiers used in the working object were Dmel_-prefixed IDs. Therefore, the mitochondrial threshold was effectively non-informative in the supplied run.

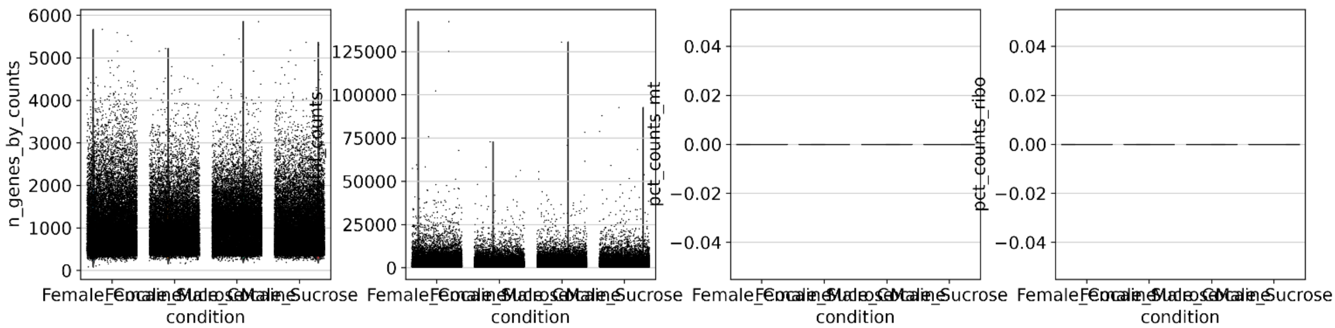

Figure 1. Supplied QC distributions across experimental conditions. The mitochondrial and ribosomal panels are flat at zero in the executed output, indicating that those feature classes were not successfully recognized.

3.3 Normalization, highly variable genes and scaling

Raw counts were stored in an AnnData layer, normalized to a target library size of 10,000 counts per cell, log1p-transformed, and 2,000 highly variable genes were selected using sample as the batch key. Total counts and mitochondrial percentage were regressed out and expression values were scaled with a maximum value of 10.

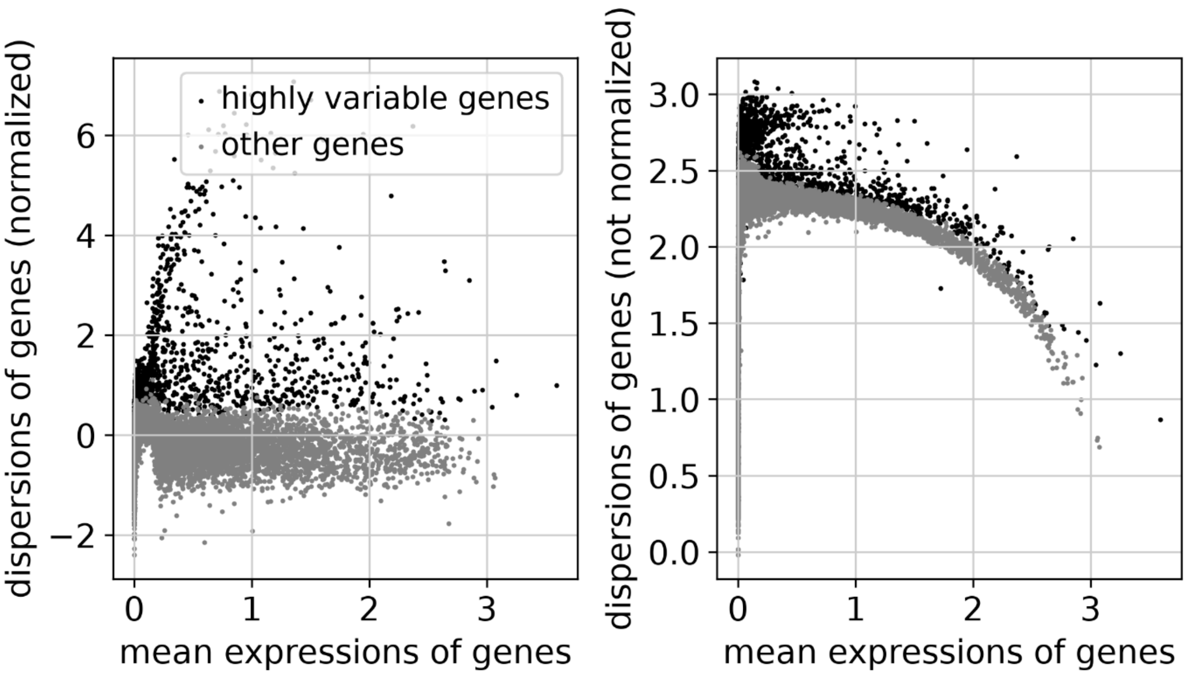

Figure 2. Highly variable gene selection used to define the 2,000-gene feature space for dimensionality reduction.

3.4 Dimensionality reduction and clustering

PCA was computed on the selected highly variable genes. Thirty principal components were used to construct a 15-nearest-neighbor graph, followed by UMAP. Leiden clustering was performed at resolution 0.8, as required by the project guide. The supplied UMAP contains cluster labels 0–13, corresponding to 14 clusters.

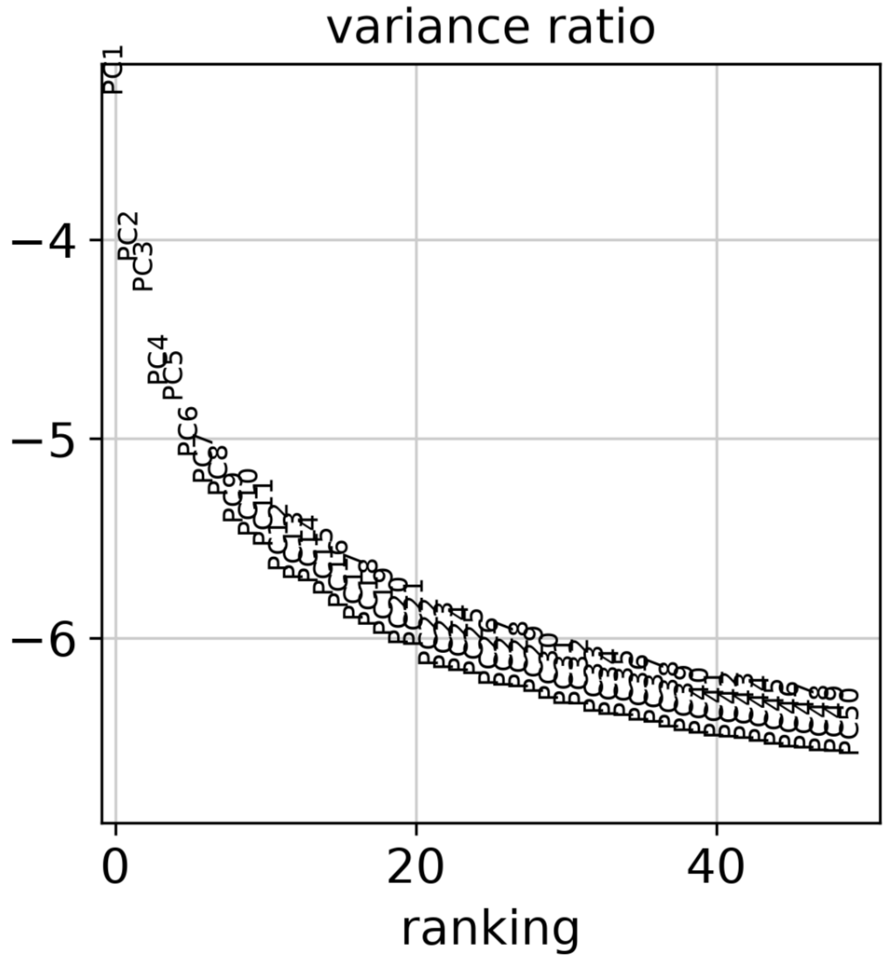

Figure 3. PCA variance-ratio plot for the first 50 principal components.

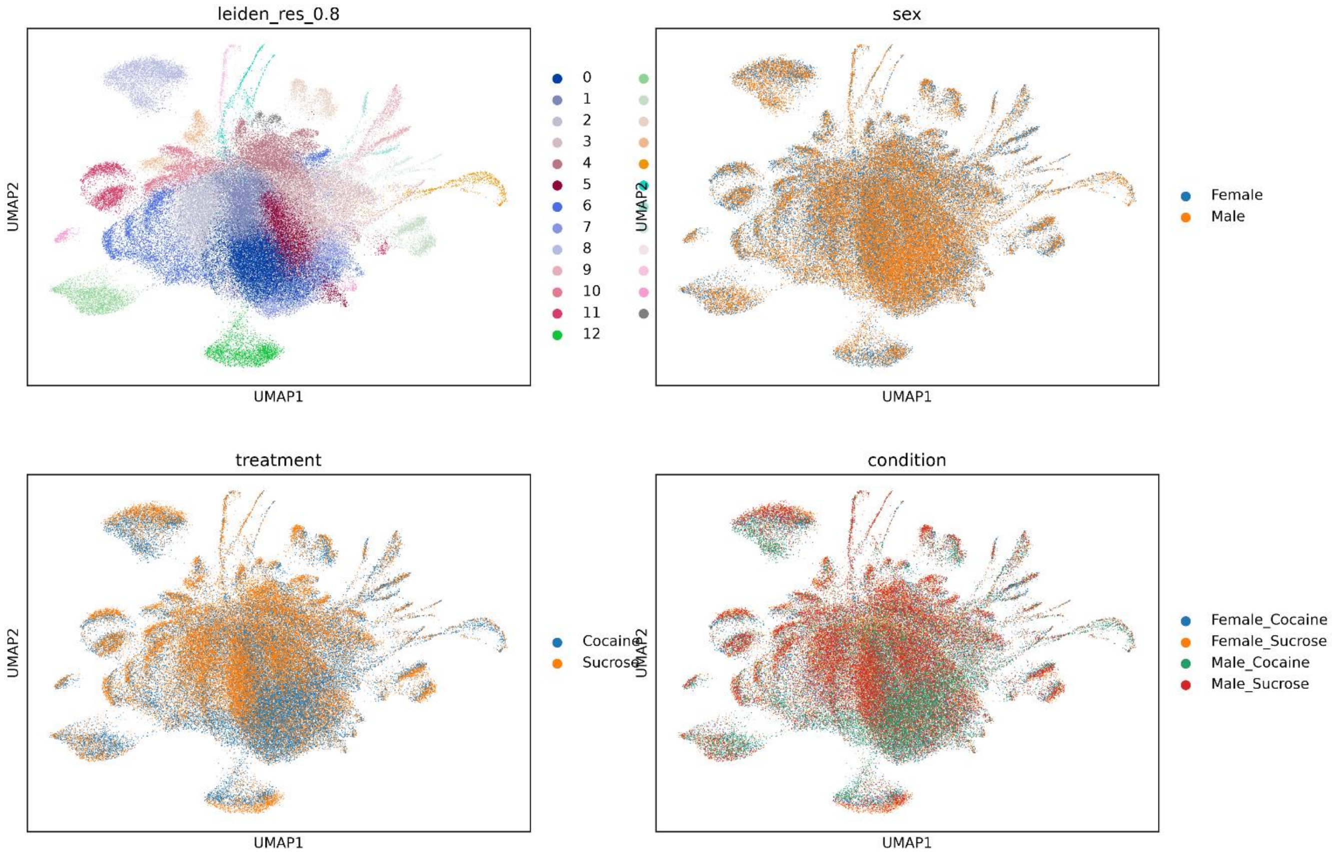

Figure 4. UMAP of the supplied analysis showing 14 Leiden clusters and distributions by sex, treatment and combined condition.

3.5 Cluster marker analysis and annotation

Cluster marker genes were ranked using the Wilcoxon test. The supplied marker plot shows the top three genes per cluster. However, the notebook attempted to map canonical markers such as repo, elav, ey, Fas2, Gad1, VGlut, ple, SerT and Tdc2, but none were found in the working feature names or raw object. Therefore, the analysis did not establish a defensible cell-type annotation from the supplied results.

Figure 5. Top three cluster marker genes from the supplied rank_genes_groups analysis. Gene identifiers remain Dmel_-prefixed, preventing direct canonical-marker annotation.

3.6 Differential expression and sex comparison

Male and female cells were analyzed separately for Cocaine versus Sucrose using Scanpy's Wilcoxon rank-sum procedure. Significant genes were defined exactly as specified by the guide: absolute log2 fold change >1.0 and adjusted P<0.05. The exported tables contain 13,141 tested genes per sex.

Sex	Significant	Up	Down	Shared with other sex
Male	104	88	16	37
Female	183	162	21	37

3.7 Pathway enrichment

The notebook used GSEApy/Enrichr for male DE genes after stripping the Dmel_ prefix. It selected available Drosophila libraries containing GO or WikiPathways terms. The supplied pathway figures display the top terms regardless of statistical significance (plot cutoff forced to 1.0). Thus these plots should be interpreted as ranked enrichment trends, not as evidence that the displayed terms are statistically enriched at FDR<0.05.

4. Results

4.1 Data retention and QC

The eight datasets contained 88,991 cells. Only 11 cells failed the ≥200-gene criterion, leaving 88,980 cells. Gene filtering removed 4,340 features, reducing the feature space from 17,481 to 13,141 genes. The resulting retention of 99.99% of cells indicates that the executed thresholds were permissive, although the missing mitochondrial/ribosomal annotation means the QC assessment was incomplete.

4.2 Atlas reconstruction

The reference study reported 36 stable clusters at resolution 0.8, whereas the supplied Scanpy workflow produced 14 clusters. The UMAP nevertheless shows structured transcriptional heterogeneity and broadly overlapping distributions of sex and treatment, rather than a simple separation of all cocaine-treated cells from all controls. This is qualitatively compatible with the idea that treatment effects are distributed across heterogeneous neural populations. However, the cluster count is substantially below the project target, so the current atlas cannot be considered a successful replication of the reference cellular resolution.

4.3 Cluster markers and cell-type identification

The top-marker dotplot demonstrates cluster-specific expression programs, but the biological identity of the clusters remains unresolved. The failure to recover canonical markers is a key limitation: it prevents direct testing of whether Kenyon-cell, glial, dopaminergic, cholinergic, GABAergic, glutamatergic, serotonergic, or octopaminergic populations are represented in the expected manner. Accordingly, claims about selective vulnerability of Kenyon cells or glia cannot be made from this marker output alone.

4.5 Leading differential-expression signals

Male — strongest positive log2FC

Gene ID	log2FC	Adjusted P
Dmel_CG14645	8.745	4.30e-04
Dmel_CG2985	7.663	1.24e-85

Dmel_CG11129	5.423	2.46e-214
Dmel_CG2979	5.254	3.61e-31
Dmel_CG34220	4.332	2.77e-08

Male — strongest negative log2FC

Gene ID	log2FC	Adjusted P
Dmel_CR32665	-9.018	0 (underflow in exported table)
Dmel_CR32777	-7.878	0 (underflow in exported table)
Dmel_CG32580	-2.626	7.44e-07
Dmel_CR45479	-2.308	9.83e-03
Dmel_CG4784	-2.251	1.63e-04

Female — strongest positive log2FC

Gene ID	log2FC	Adjusted P
Dmel_CG2985	6.194	1.75e-07
Dmel_CR33682	3.871	4.15e-204
Dmel_CG2979	3.656	4.54e-03
Dmel_CR45845	3.594	5.97e-03
Dmel_CG11129	3.473	7.54e-10

Female — strongest negative log2FC

Gene ID	log2FC	Adjusted P
Dmel_CR32665	-9.091	0 (underflow in exported table)
Dmel_CG13762	-7.647	6.92e-03
Dmel_CR32777	-7.563	0 (underflow in exported table)
Dmel_CG13057	-2.917	7.88e-03
Dmel_CG3466	-2.668	1.77e-03

4.6 Pathway enrichment

The male pathway output contains terms associated with neurotransmitter receptor complexes, acetylcholine-gated channel complexes, terminal boutons, plasma membrane components, phagocytic vesicles, peroxisomal membrane, mitochondrial ATP synthase, AMPA glutamate receptor complex, glycerolipid metabolism, amino-acid catabolism, glutamate metabolism, glucose homeostasis and related processes. These categories are biologically compatible with neuronal signaling and metabolic remodeling, but the supplied plot shows −log10(adjusted P) values well below the 1.30 threshold corresponding to adjusted P=0.05. Therefore, none of the displayed pathways should be reported as statistically significant enrichment.

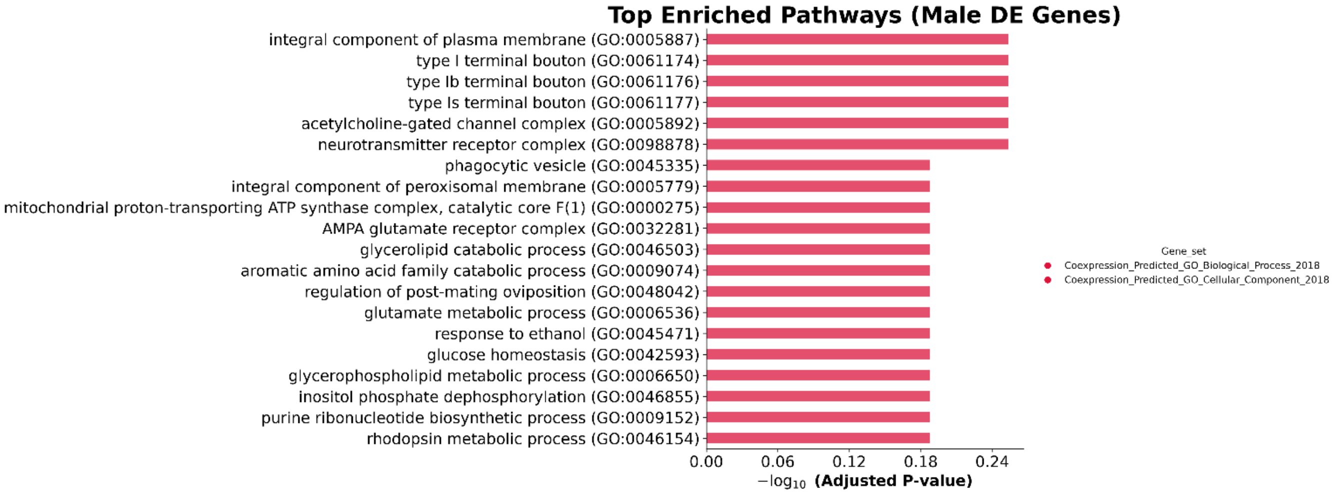

Figure 7. Supplied male pathway enrichment bar plot. The plot ranks terms but uses a forced plotting cutoff; displayed terms do not cross the adjusted-P<0.05 significance threshold.

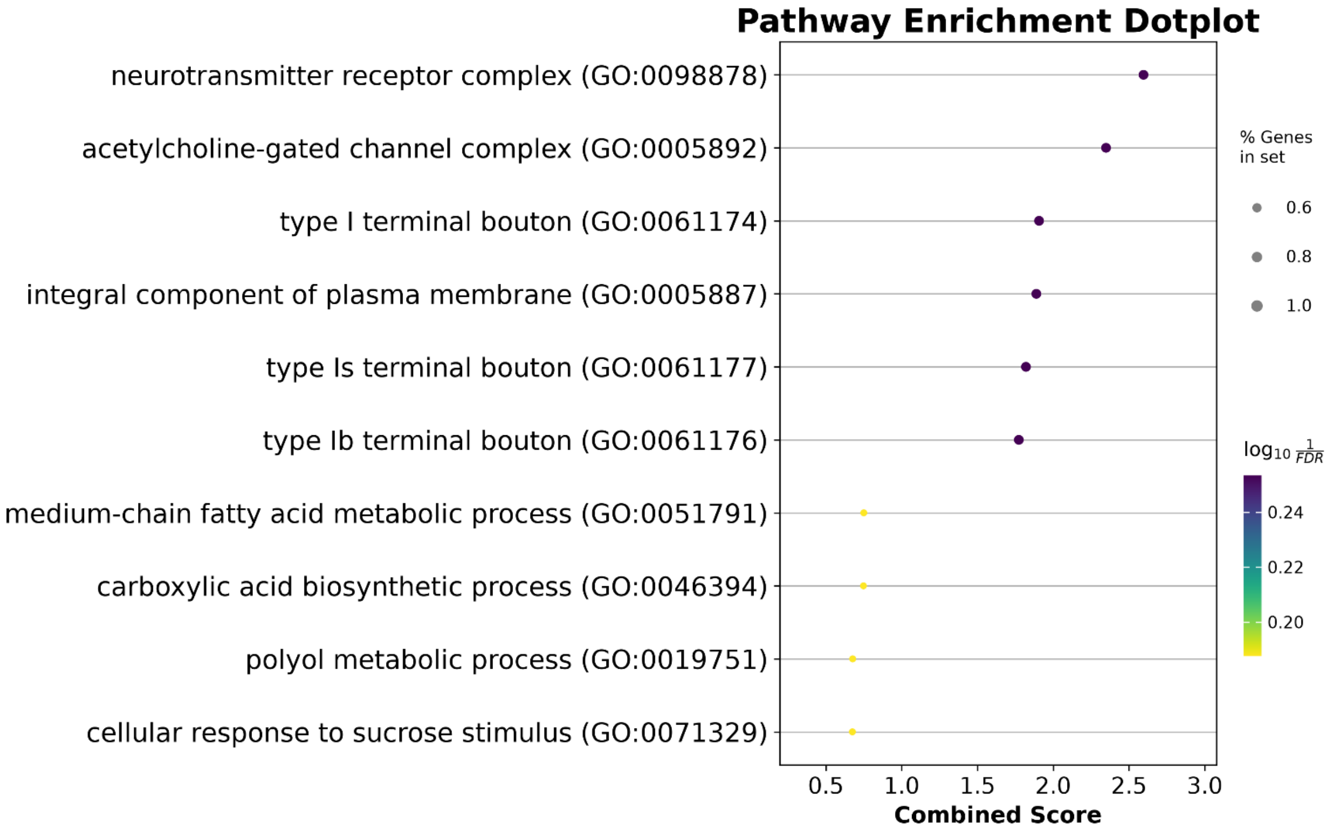

Figure 8. Supplied pathway enrichment dotplot. Dot size represents the percentage of genes in the set and color represents the plotted enrichment statistic.

5. Discussion

5.1 Reproducibility relative to the reference study

The analysis reproduces several structural elements of the intended workflow: all eight experimental groups were loaded, merged into a single AnnData object, filtered, normalized, reduced by

PCA/UMAP, clustered with Leiden at resolution 0.8, and subjected to sex-stratified cocaine-versus-sucrose differential expression. These steps establish a valid framework for the project.

However, the current output diverges materially from the reference atlas. The reference study reported 36 clusters at resolution 0.8, whereas the supplied UMAP contains 14. The reference study also reported 691 DEGs in males and 322 in females using its published analytical workflow, whereas the supplied output contains 104 and 183 significant genes, respectively. These discrepancies are not merely numerical: they directly affect the ability to resolve rare neuronal/glial populations and to attribute cocaine responses to specific cell types.

5.2 Likely analytical contributors to the discrepancy

- Gene identifiers were retained as Dmel_-prefixed IDs, preventing canonical marker detection and weakening biological annotation.
- Mitochondrial and ribosomal QC annotations returned zero percentages, so those QC dimensions were not actually used to discriminate low-quality cells.
- The cell-level Wilcoxon DE analysis does not explicitly model the two biological replicates per sex and treatment; this can inflate apparent evidence when cells are treated as independent observations.
- The workflow uses log-normalized/scaled expression and regression rather than the SCTransform-based normalization/integration strategy used in the reference analysis.
- The present report analyzes whole-sex DE rather than the reference study's cluster-specific cocaine responses, which are central to identifying glial and Kenyon-cell sensitivity.
- The pathway analysis was visualized with a forced cutoff of 1.0, so visually prominent terms are not necessarily statistically enriched.

5.3 Biological interpretation

Within the limits of the current pipeline, cocaine exposure is associated with substantial transcriptional remodeling in both sexes, with a predominantly positive log2FC distribution among significant genes. The modest overlap of 37 significant genes between male and female sets indicates that much of the detected response is sex-specific under the present statistical framework. The male pathway ranking also contains neuronal signaling and membrane-associated categories alongside metabolic processes, suggesting that cocaine-associated changes may span synaptic machinery and cellular metabolism. Because the pathway results are not FDR-significant and cell-type annotation is incomplete, these observations should be described as hypotheses rather than validated mechanisms.

6. Conclusion

The supplied Scanpy analysis successfully establishes a reproducible single-cell processing framework for the eight Drosophila brain samples and demonstrates cocaine-associated transcriptional changes in both sexes. Nevertheless, it does not yet reproduce the reference atlas or its reported male-biased cocaine response. The principal outputs are 88,980 post-filter cells, 13,141 retained genes, 14 Leiden clusters, 104 male significant DEGs and 183 female significant DEGs, with 37 genes shared between sexes. The most important conclusion is therefore not that females are more cocaine-responsive, but that the current pipeline produces a sex-specific result that conflicts with the reference study and requires methodological reconciliation. In particular, the failed mitochondrial/ribosomal annotation, absent canonical marker mapping, lower-than-target cluster resolution, and non-replicate-aware DE analysis limit biological interpretation. The pathway plots provide useful hypotheses around neuronal signaling and metabolism, but none of the displayed terms meets the stated adjusted-P significance criterion. A revised analysis addressing these issues would be necessary before making publication-level claims about Kenyon cells, glia, or sexual dimorphism in cocaine-induced transcriptional responses.

7. References

Baker BM, Mokashi SS, Shankar V, Hatfield JS, Hannah RC, Mackay TFC, Anholt RRH. The Drosophila brain on cocaine at single-cell resolution. Genome Research. 2021;31(10):1927–1938. The study used GEO accession GSE152495 and reported 36 clusters and stronger cocaine-associated transcriptional responses in males.

Project 2 Guide. Drosophila Brain Cocaine Response: Single-Cell RNA-Seq Analysis. Supplied course/project document.

NCBI Gene Expression Omnibus. GSE152495: The Drosophila Brain on Cocaine at Single Cell Resolution.

Appendix A. AI Usage Disclosure

AI assistance was used to organise the supplied Project 2 guide and supplied computational figures into a structured academic report, improve wording, and identify discrepancies between the guide's expected deliverables and the supplied outputs. The AI model used in this drafting step was ChatGPT (GPT-5.6 Luna).

Appendix B. Supplied Output Inventory

Figure1_QC_metrics.png
Figure2_UMAP_clusters.png
Figure3_top_cluster_markers.png
Figure4_pathway_dotplot.png
Figure5_pathway_enrichment.png
Figure6_HVG_selection.png
Figure7_PCA_variance.png